

University of Dhaka



Assignment - 2

Course Title: Advanced Econometrics-II
Course Code: 408

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Question:

A data analyst estimated an ARMA (2,1) model for changes in the clothing industry's profits. However, the true model is actually an ARMA (2,2). Use simulations to show that the data analyst will estimate a biased model.

Solution:

Let's assume an ARMA (2,2) model is,

$$y_t = 0.4y_{t-1} + 0.3y_{t-2} + \varepsilon_t + 0.5\varepsilon_{t-1} + 0.3\varepsilon_{t-2}$$

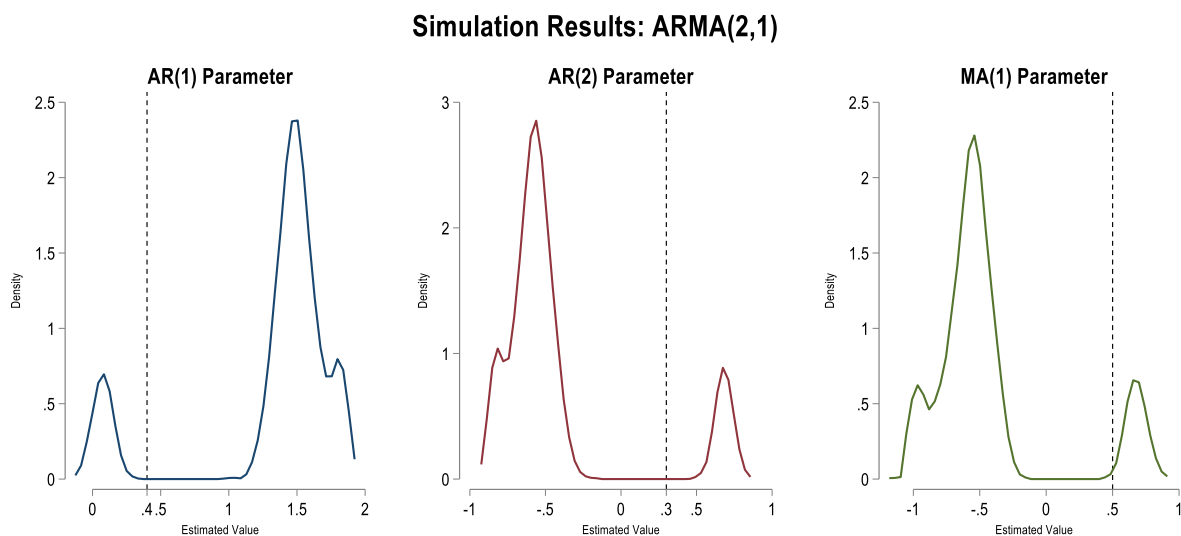
Where the coefficient of AR (1) = 0.4

coefficient of AR (2) = 0.3

coefficient of MA (1) = 0.5

coefficient of MA (2) = 0.3

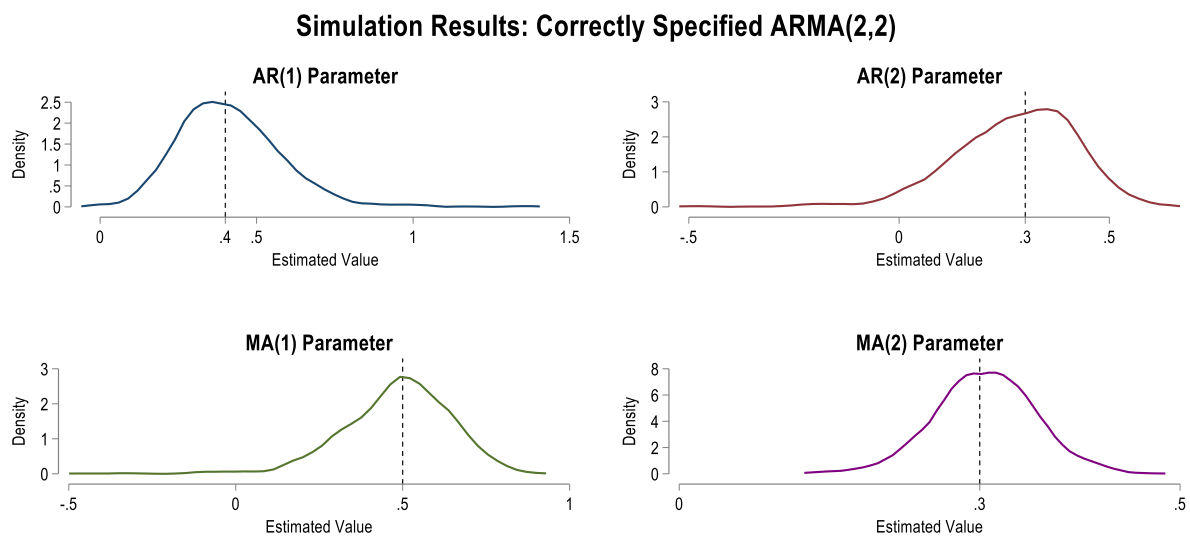
Now if a data analyst estimated an ARMA (2,1) model then the estimated coefficients will be biased downward or upward. To show this we will run a simulation with 1000 samples. With each sample we will estimate the coefficient, and then we will compare the expected value or mean of coefficients with the true parameters.



Black dashed line represents the true parameter

Coefficients	True value	Estimated Value	Biased
AR (1)	0.4	1.32	Upward by 0.92 points
AR (2)	0.3	-0.42	Downward by 0.72 points
MA (1)	0.5	-0.43	Downward by 0.93 points

From the diagram we can see that the estimated values exhibit multimodal distribution and does not converge to the single true parameters. The AR (1) parameter is estimated biased upward and both the AR (2) and MA (1) parameters are estimated biased downward.



Black dashed line represents the true parameter

Now if the data analyst had correctly specified the ARMA (2,2) model then the simulation with 1000 samples would yield similar estimated values close to true parameters and their distribution would follow the same true parameters which we can see from diagram for correctly specified ARMA (2,2) model.

In conclusion, a miss specified ARMA (2,1) model would yield biased coefficients for each value when the true model is ARMA (2,2).

```

1  cd "F:\ECON 408\Practice do file"
2
3
4  * (1) Biased ARMA(2,1) Model Simulation
5  *-----
6
7  clear all
8  set seed 1000
9
10 capture postclose buffer
11 postfile buffer beta_L1 beta_L2 alpha_L1 using arma22_misspecified.dta, replace
12
13 forvalues i = 1/1000 {
14     qui drop _all
15     qui set obs 500
16     qui gen time = _n
17     qui tsset time
18
19     qui gen double e = rnormal()
20     qui gen double y = e in 1
21     qui replace y = 0.4*L1.y + e + 0.5*L1.e in 2
22     qui replace y = 0.4*L1.y + 0.3*L2.y + e + 0.5*L1.e + 0.3*L2.e in 3/L
23     qui arima y, ar(1/2) ma(1)
24     post buffer (_b[ARMA:L1.ar]) (_b[ARMA:L2.ar]) (_b[ARMA:L1.ma])
25
26 }
27
28 postclose buffer
29 use arma22_misspecified.dta,clear
30
31 sum beta_L1 beta_L2 alpha_L1
32
33 set scheme white_tableau
34 graph set window fontface "Arial Narrow"
35
36 * AR(1) Plot
37 kdensity beta_L1, lcolor(navy) lpattern(solid) lwidth(medthick) ///
38     xline(0.4, lcolor(black) lpattern(shortdash)) ///
39     name(g1, replace) title("{bf:AR(1) Parameter}") ///
40     xtitle("Estimated Value") ///
41     xlabel(0 0.4 0.5 1 1.5 2 ,labsize(medium) nogrid) ///
42     ylabel(, labsize(medium) nogrid) note(" ")
43
44 * AR(2) Plot
45 kdensity beta_L2, lcolor(maroon) lpattern(solid) lwidth(medthick) ///
46     xline(0.3, lcolor(black) lpattern(shortdash)) ///
47     name(g2, replace) title("{bf:AR(2) Parameter}") ///
48     xtitle("Estimated Value") ///
49     xlabel(-1 -0.5 0 0.3 0.5 1 ,labsize(medium) nogrid) ///
50     ylabel(, labsize(medium) nogrid) note(" ")
51
52 * MA(1) Plot
53 kdensity alpha_L1, lcolor(forest_green) lpattern(solid) lwidth(medthick) ///
54     xline(0.5, lcolor(black) lpattern(shortdash)) ///
55     name(g3, replace) title("{bf:MA(1) Parameter}") ///
56     xtitle("Estimated Value") xlabel(,labsize(medium) nogrid) ///
57     note(" ") ylabel(, labsize(medium) nogrid)
58
59 * Combine Graph
60 graph combine g1 g2 g3, ///
61     rows(1) title("{bf:Simulation Results: ARMA(2,1)}") ///
62     xsize(8) ysize(4) ///
63     note("Black dashed line represents the true parameter") ///

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64     name(Biased, replace)
65
66 graph export Biased_arma.emf, name(Biased) replace
67
68
69
70 * (2) Unbiased Correct ARMA(2,2) Model Simulation
71 *-----
72
73 clear all
74 set seed 1000
75
76 capture postclose buffer
77 postfile buffer beta_L1 beta_L2 alpha_L1 alpha_L2 using arma22_correct_model.dta,
  replace
78
79 forvalues i = 1/1000 {
80     qui drop _all
81     qui set obs 500
82     qui gen time = _n
83     qui tsset time
84
85     qui gen double e = rnormal()
86     qui gen double y = e in 1
87     qui replace y = 0.4*L1.y + e + 0.5*L1.e in 2
88     qui replace y = 0.4*L1.y + 0.3*L2.y + e + 0.5*L1.e + 0.3*L2.e in 3/L
89     arima y, ar(1/2) ma(1/2)
90     matlist e(b)
91     post buffer (_b[ARMA:L.ar]) (_b[ARMA:L2.ar]) (_b[ARMA:L.ma]) (_b[ARMA:L2.ma])
92
93 }
94
95 postclose buffer
96 use arma22_correct_model.dta,clear
97
98 sum beta_L1 beta_L2 alpha_L1 alpha_L2
99
100 set scheme white_tableau
101 graph set window fontface "Arial Narrow"
102
103
104 * AR(1) Plot
105 kdensity beta_L1, lcolor(navy) lpattern(solid) lwidth(medthick) ///
106     xline(0.4, lcolor(black) lpattern(shortdash)) ///
107     name(g1, replace) title("{bf:AR(1) Parameter}") ///
108     xtitle("Estimated Value", size(medium)) ytitle(, size(medium)) note(" ") ///
109     xlabel(0 0.4 0.5 1 1.5, labsize(medium) nogrid) ylabel(, labsize(medium)
nogrid)
110
111 * AR(2) Plot
112 kdensity beta_L2, lcolor(maroon) lpattern(solid) lwidth(medthick) ///
113     xline(0.3, lcolor(black) lpattern(shortdash)) ///
114     name(g2, replace) title("{bf:AR(2) Parameter}") ///
115     xtitle("Estimated Value", size(medium)) ytitle(,size(medium)) note(" ") ///
116     xlabel(-0.5 0 0.3 0.5, labsize(medium) nogrid) ylabel(, labsize(medium)
nogrid)
117
118 * MA(1) Plot
119 kdensity alpha_L1, lcolor(forest_green) lpattern(solid) lwidth(medthick) ///
120     xline(0.5, lcolor(black) lpattern(shortdash)) ///
121     name(g3, replace) title("{bf:MA(1) Parameter}") ///
122     xtitle("Estimated Value", size(medium)) ytitle(,size(medium)) note(" ") ///
123     xlabel(-0.5 0 0.5 1, labsize(medium) nogrid) ylabel(, labsize(medium) nogrid)

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124
125 * MA(2) Plot
126 kdensity alpha_L2, lcolor(purple) lpattern(solid) lwidth(medthick) ///
127     xline(0.3, lcolor(black) lpattern(shortdash)) ///
128     name(g4, replace) title("{bf:MA(2) Parameter}") ///
129     xtitle("Estimated Value",size(medium)) ytitle(,size(medium)) note(" ") ///
130     xlabel(0 0.3 0.5, labsize(medium) nogrid) ylabel(, labsize(medium) nogrid)
131
132 * Combine into 2x2 Grid
133 graph combine g1 g2 g3 g4, ///
134     rows(2) cols(2) ///
135     title("{bf:Simulation Results: Correctly Specified ARMA(2,2)}") ///
136     note("Black dashed line represents the true parameter", size(small)) ///
137     xsize(8) ysize(4) name(CorrectModel, replace)
138
139
140 graph export ARMA_Correct.emf, name(CorrectModel) replace
141
142
143
```